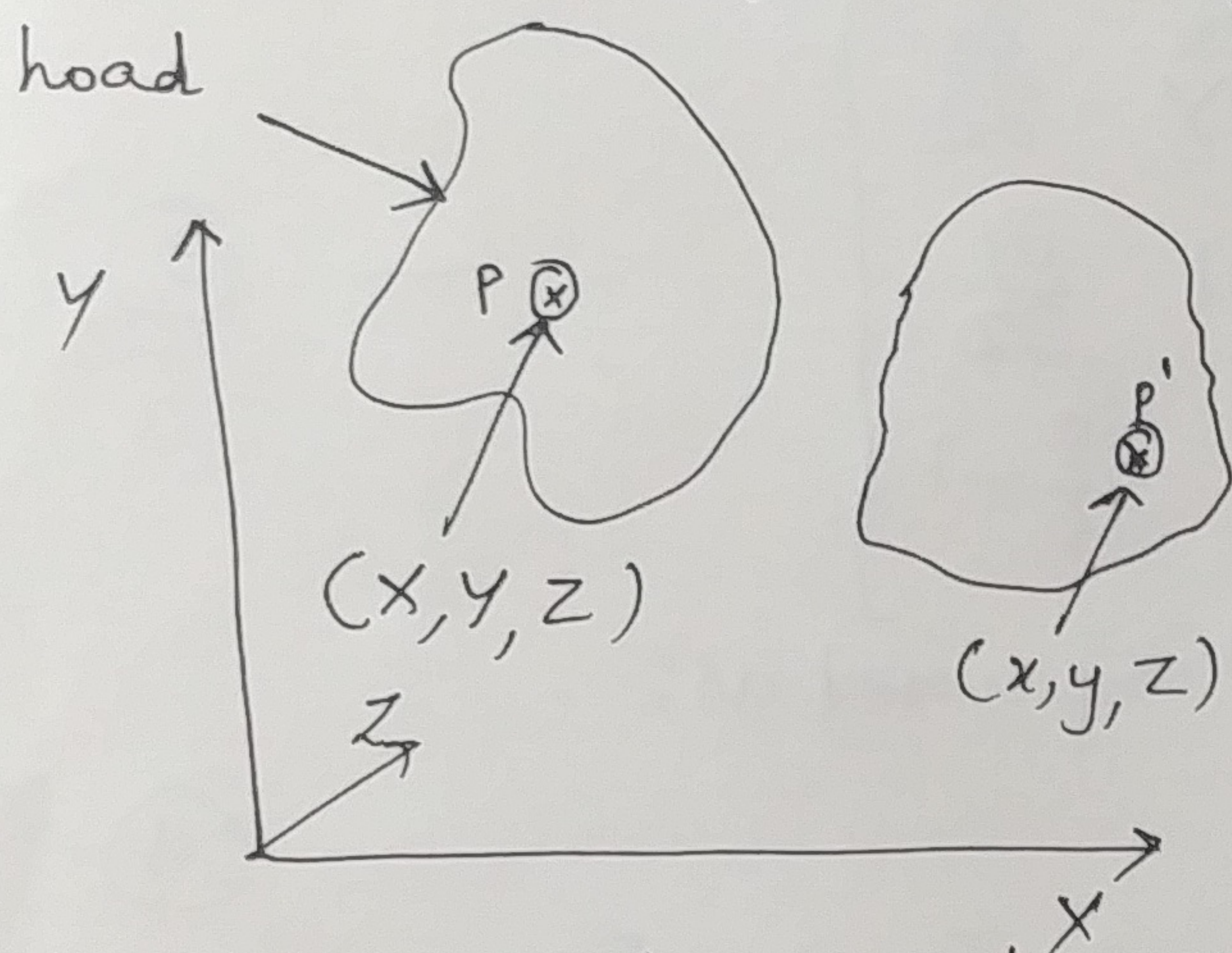


Non-linear solid Mechanics - Constitutive relations

Soft tissues, in general follow the principles of non-linear mechanics - hence non-linear stress strain relations and corresponding constitutive relations need to be used. In non-linear mechanics, the generalized principles of deformation, strain and stress need to be utilized. Considering the deformation of an arbitrary body;



$P \rightarrow (X, Y, Z) \rightarrow$ old position of an arbitrary point inside the body

$P' \rightarrow (x, y, z) \rightarrow$ final position (post deformation) of the same point inside the body

3D deformation and point displacement

The displacements in (X, Y, Z) directions can be expressed as

$$u_x = x - X ; \quad u_y = y - Y ; \quad u_z = z - Z \quad \text{--- ①}$$

where $x = f(X, Y, Z)$, $y = f(X, Y, Z)$, $z = f(X, Y, Z)$

Hence, the displacement gradients are;

$$\frac{\partial u_x}{\partial X} = \frac{\partial x}{\partial X} - 1, \quad \frac{\partial u_y}{\partial Y} = \frac{\partial y}{\partial Y} - 1, \quad \frac{\partial u_z}{\partial Z} = \frac{\partial z}{\partial Z} - 1 \quad \text{--- ②}$$

Similarly,

$$\frac{\partial u_x}{\partial y} = \frac{\partial x}{\partial y}, \quad \frac{\partial u_x}{\partial z} = \frac{\partial x}{\partial z}, \quad \frac{\partial u_y}{\partial x} = \frac{\partial y}{\partial x}, \quad \frac{\partial u_y}{\partial z} = \frac{\partial y}{\partial z}$$

$$\frac{\partial u_z}{\partial x} = \frac{\partial z}{\partial x}, \quad \frac{\partial u_z}{\partial y} = \frac{\partial z}{\partial y}$$

because (x, y, z) are independent of each other
 $\left\{ \begin{array}{l} \frac{\partial x}{\partial y} = 0, \frac{\partial y}{\partial z} = 0 \dots \end{array} \right\}$

Hence, the displacement gradient can be written in matrix form as;

$$[H] = \begin{bmatrix} \frac{\partial x}{\partial X} - 1 & \frac{\partial x}{\partial Y} & \frac{\partial x}{\partial Z} \\ \frac{\partial y}{\partial X} & \frac{\partial y}{\partial Y} - 1 & \frac{\partial y}{\partial Z} \\ \frac{\partial z}{\partial X} & \frac{\partial z}{\partial Y} & \frac{\partial z}{\partial Z} - 1 \end{bmatrix} \quad \text{--- (3)}$$

Equation (3) can also be expressed as;

$$[H] = [F] + [I] \quad \text{--- (4)}$$

where $[I] = \text{Identity matrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

and; $[F] = \text{deformation gradient} = \begin{bmatrix} \frac{\partial x}{\partial X} & \frac{\partial x}{\partial Y} & \frac{\partial x}{\partial Z} \\ \frac{\partial y}{\partial X} & \frac{\partial y}{\partial Y} & \frac{\partial y}{\partial Z} \\ \frac{\partial z}{\partial X} & \frac{\partial z}{\partial Y} & \frac{\partial z}{\partial Z} \end{bmatrix} \quad \text{--- (5)}$

In equation (5), the deformation gradient matrix indicates the relative displacements of points in the body.

The generalized expression for strain (Green's strain) can be expressed as;

$$[E] = \frac{1}{2} [[F]^T [F] - [I]] \quad \text{--- (6)}$$

where $[F]$ is the deformation gradient matrix and $[I]$ is the identity matrix. Equation (6) is applicable for both linear and non-linear problems. The equation for linearized strains can be written as;

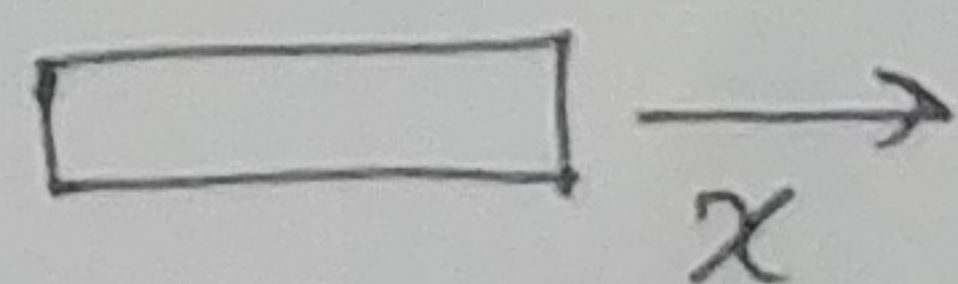
$$[\epsilon] = \frac{1}{2} [[F]^T + [F] - 2[I]] \quad \text{--- (7)}$$

It should be mentioned that equation (7) is applicable only for the case of small strains, while equation (6) is applicable for small and large strains. Results from equation (6) converge to the values obtained from equation (7), in the case of small strains.

Case 1 Simple Stretching in one-direction

The relation between the initial and final coordinates of a point in the above case of loading would be;

$$x = \Lambda X, \quad y = Y, \quad z = Z$$



where Λ = stretch ratio

The deformation gradient matrix in this case can be written as;

$$[F] = \begin{bmatrix} \Lambda & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad [F]^T = \begin{bmatrix} \Lambda & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[F]^T [F] = \begin{bmatrix} \Lambda^2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(generalized)

$$\text{Hence } \bar{E} = \frac{1}{2} [[F]^T [F] - I] = \frac{1}{2} \begin{bmatrix} \Lambda^2 - 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The strain component $E_{11} = \frac{1}{2}(\Lambda^2 - 1)$

$$\text{For } \Lambda = 1.1 \text{ (10\% stretching)} \quad \bar{E}_{11} = \frac{1}{2}(1.21 - 1) = \underline{\underline{0.105}}$$

$$\text{lineal strain } [E] = \frac{1}{2} [[F] + [F]^T - 2[I]]$$

$$= \frac{1}{2} \left[\begin{bmatrix} 2\Lambda & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \right]$$

$$[E] = \frac{1}{2} \begin{bmatrix} 2(\Lambda - 1) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \underline{\underline{\Lambda - 1}}$$

$$E_{11} \text{ corresponding to } (\Lambda = 1.1) = \underline{\underline{0.1}} \simeq \bar{E}_{11}$$

Hence, it can be observed that the difference between the generalized and linear strains is minimal.

For $\Lambda = 2$ (100% stretching)

$$\bar{\epsilon}_{11} = \frac{1}{2}(4-1) = 1.5, \text{ and } \epsilon_{11} = 2-1 = 1$$

Hence, the difference between $\bar{\epsilon}_{11}$ and ϵ_{11} keeps increasing for larger strains (stretch ratios).

Case 2 Single plane shear loading

$$x = X + \alpha Y, \quad y = Y, \quad z = Z$$

$$\text{Deformation gradient, } [F] = \begin{bmatrix} 1 & \alpha & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{The transpose matrix } [F]^T = \begin{bmatrix} 1 & 0 & 0 \\ \alpha & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[F]^T [F] - [I] = \begin{bmatrix} 1 & \alpha & 0 \\ \alpha & 1+\alpha^2 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & \alpha & 0 \\ \alpha & \alpha^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence, the generalized green's strain matrix is expressed as ;

$$[\bar{\epsilon}] = \begin{bmatrix} 0 & \frac{\alpha}{2} & 0 \\ \frac{\alpha}{2} & \frac{\alpha^2}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The generalized strain is expressed as;

$$[\varepsilon] = \frac{1}{2} \left[\begin{bmatrix} 1 & \alpha & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ \alpha & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \right]$$

$$[\varepsilon] = \frac{1}{2} \begin{bmatrix} 0 & \alpha & 0 \\ \alpha & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

⑨

$$\varepsilon_{22} = \frac{\alpha^2}{2}, \text{ where } \varepsilon_{22} = 0$$

Hence, the generalized strain matrix $[E]$ is able to capture normal strain in Y direction corresponding to shear loading in X-Y plane, while linearized strain models fail to capture this. It should be noted that biological organs are subjected to multiple plane shear in many cases.

Case 3

Multiple plane shear

$$x = \alpha X + \beta Y, \quad y = \beta X + \gamma Y, \quad z = \delta X + \alpha Y + \beta Z$$

In this case,

$$[F] = \begin{bmatrix} \alpha & \beta & 0 \\ \beta & \gamma & 0 \\ \delta & \alpha & \beta \end{bmatrix}$$

$$F^T = \begin{bmatrix} \alpha & \beta & \delta \\ \beta & \gamma & \alpha \\ 0 & 0 & \beta \end{bmatrix}$$

Cauchy and Piola-Kirchhoff stress

The concepts of Cauchy stress and piola-Kirchhoff stress are widely utilized in the calculation of mechanical behavior of materials. Cauchy stress representations are widely utilized in computations, and the stress is calculated based on varying areas of cross-section during the process of deformation. On the other hand, piola-Kirchhoff stress is based on the initial area of cross-section, and hence, widely used in experimental characterization of materials. Therefore the Cauchy stress (σ) and piola-Kirchhoff stress are linked through the representations of deformation gradient matrix, $[F]$.

The equation for Cauchy stress on a deformable body is given as;

$$[\sigma] = \frac{1}{|F|} [F] [S] [F]^T \quad \text{--- (10)}$$

In the above equation, $[S]$ represents the second piola-Kirchhoff matrix/tensor. Representation $[S]$ indicates second piola-Kirchhoff stress. The second piola-Kirchhoff stress (p-K) is linked to the strain energy of the material.

The strain energy is defined as the energy stored in a body due to deformation. The second-(p-k) stress is linked to the strain energy using the relation;

$$[S] = \left[\frac{\partial W}{\partial E} \right], \quad \text{--- (11)}$$

where $[E]$ represents the generalized Green's strain. The strain energy of the body can be expressed in terms of the non-linear deformations / strains as per the case of loading.

There are numerous analytical expressions for strain energy. A few examples are provided below.

$$W = C_1 E_{11}^2 + C_2 E_{22}^2 + C_3 E_{33}^2 + C_4 E_{12} E_{21} + C_5 E_{23} E_{32} + C_6 E_{31} E_{13}$$

$$W = \frac{1}{2} c [e^Q - 1] \quad (\text{Fung's model}) \quad \text{--- (13)}$$

where

$$Q = f(E_{ij})$$

In the above equation, C_n represents the material-dependent constants; that can be experimentally determined from tissue/material - deformation experiments. It should be emphasized that the equation for W is not unique and depends on the tissue - eventually depending on the microstructure and protein fiber arrangements.